Vision-Language-Action Model and Diffusion Policy Switching Enables Dexterous Control of an Anthropomorphic Hand

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Keywords:

- Vision-Language-Action (VLA) models
- Diffusion policy for robot control
- Dexterous manipulation, anthropomorphic hand
- Hybrid high-level / low-level control

Research Area and Background (1/2)

- Dexterous manipulation requires:
    - Advanced task planning
    - Wrist trajectory generation
    - Contextual grasp selection
    - Precise multi-finger control
- Large Vision-Language-Action (VLA) models:
    - Translate language commands to robot actions
    - Trained on large-scale manipulation datasets (e.g., Open-X-Embodiment, DROID, BridgeData v2)
- Examples: RT-1, RT-2, Octo, openVLA

Research Area and Background (2/2)

- Existing VLA deployments:
  - Primarily on 1 DoF pinch grippers
  - Limited evidence on multi-fingered, anthropomorphic hands
- Diffusion policy models:
  - Well-suited for robot motion learning and planning
  - Can generate smooth, precise motor commands
  - Capture contextual information from sensor inputs
- Prior diffusion work includes multi-fingered hands, but typically task-specific

Goal of the Research

- Develop a **hybrid control framework** combining:
  - Fine-tuned openVLA model for **language-conditioned high-level planning**
  - Diffusion policy for **low-level dexterous grasping control**
- Enable **autonomous pick-and-place** with:
  - Language-specified target object and placement location
  - Multi-fingered anthropomorphic hand (ADAPT Hand 2)
- Demonstrate **event-based switching** between VLA and diffusion policies
- Achieve higher success rates than VLA-only control

State-of-the-Art and Limitations

- **VLAs (e.g., RT-1, RT-2, Octo, openVLA):**
  - Strength: generalizable language-to-action mapping
  - Limitation: demonstrated mainly on simple grippers
  - Unclear precision for complex anthropomorphic hands
- **Visual servoing:**
  - Effective for approaching and grasping
  - Limited use of contextual/environmental information for grasp strategy
- **Diffusion policies:**
  - Strength: precise, smooth control; multi-modal behavior
  - Limitation: task- and object-specific training; limited global planning

Research Motivation

- VLAs:
  - Good at high-level planning from language
  - Struggle with precise, contact-rich dexterous manipulation
- Diffusion policies:
  - Good at local, precise control and exploiting context
  - Lack global task planning and language grounding
- **Key idea:** Combine VLA and diffusion to exploit complementary strengths
  - VLA: move hand near target from language
  - Diffusion: execute robust grasp and release
- Target platform: compliant anthropomorphic ADAPT Hand 2 (13 DoF)

Problem Definition

- Task: **Autonomous pick-and-place** with:
  - Language command: "pick [object] and place on [plate color]"
  - Multiple objects and placement locations
- Robot:
  - ADAPT Hand 2 (13 DoF) mounted on UR5 arm
  - Two RGB cameras: static (cam1) and wrist-mounted (cam2)
- Requirements:
  - Language-conditioned high-level motion
  - Precise dexterous grasping and release
  - Automatic switching between controllers
- Baseline: VLA-only control for full pick-and-place

- **VLA (openVLA, fine-tuned):**
  - Input: language command + images (cam1+cam2)
  - Output: 6D end-effector pose + scalar grasp/event signal
- **Diffusion policy:**
  - Input: images (cam1, cam2) + state
  - Output: multi-dimensional hand joint commands
- **Event signal σ:**
  - Shared scalar signal to mark key task events
  - Drives automatic switching between policies

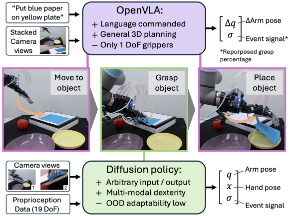

Figure: Combined VLA and diffusion policy approach for dexterous manipulation (Fig. 1).

VLA & Diffusion Switching Framework

- Use a **common event signal** σ to:
  - Detect proximity to object / plate (VLA)
  - Detect successful grasp (diffusion)
- Execution sequence:
  - VLA moves hand near object; σ rises
  - Switch to diffusion for grasping; σ rises at successful lift
  - Switch back to VLA for transport and placement
- Object-specific diffusion policies:
  - Selected via lookup table from language input

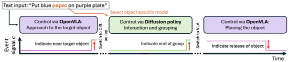

Figure: Switching between VLA and diffusion using event signal σ (Fig. 2).

Robotic Platform: ADAPT Hand 2

- Anthropomorphic hand with 13 motors
- Series elastic actuation at base joints:
  - Compliance for safe interaction
  - Physical filtering of noisy control signals
- Anatomical kinematic design and continuous soft skin
- Mounted on UR5 arm for 6-DoF positioning

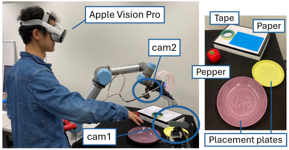

Figure: ADAPT Hand 2 with compliant joints and continuous skin (Fig. 3).

Experimental Setup and Teleoperation

- Teleoperation for data collection:
  - Apple Vision Pro hand tracking
  - Joint-to-joint mapping to UR5 + ADAPT Hand 2
- Sensors:
  - cam1: static world camera
  - cam2: wrist-mounted camera
- Objects:
  - Red pepper, tape roll, piece of paper
  - Two plates (yellow, purple) as placement targets
- Inference hardware: NVIDIA RTX 4090, ≈5 Hz control

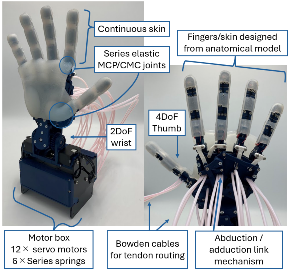

Figure: Teleoperation setup and test objects/plates (Fig. 4).

Data Collection Strategy

- **VLA dataset (Fig. 5A):**
  - Teleoperated full pick-and-place trajectories
  - Operator records event signal via controller
  - Only pre- and post-grasp segments used for training
  - Hand deliberately closed in mid-air when event signal first applied
  - 20 trials per object-placement combination
- **Diffusion dataset (Fig. 5B):**
  - Only grasping segments
  - Hand initialized $\approx$5 cm above object
  - 40 trials (tape, paper), 30 trials (pepper)
  - Includes multiple grasp strategies and failure-recovery attempts
  - Event signal non-zero only in final 10 steps of successful trials

Data Collection Illustration

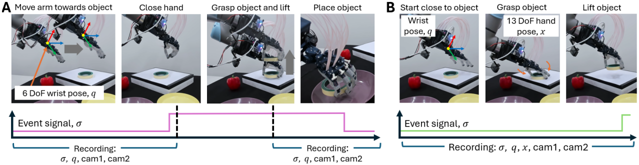

Figure: Data collection for VLA (A) and diffusion policy (B), including event signal annotation (Fig. 5).

Model Training Details

- **VLA (openVLA fine-tuning):**
  - Inputs: concatenated images from cam1 (224×144) and cam2 (224×80)
  - Images vertically concatenated into single frame
  - Default openVLA fine-tuning settings, batch size 22
  - No image augmentation (fixed camera pose and lighting)
  - Trained until action accuracy $> 95\%$ ($\approx$50k steps)
  - Trained on A100-80GB GPU
- **Diffusion policy:**
  - Inputs: images from cam1, cam2 (320×240), random crop to 288×216
  - CNN-based diffusion policy as in prior work
  - Trained for 1500 epochs on custom GPU

Experimental Design

- **VLA evaluation:**
  - Full grasping and placing with VLA-only
  - Objects: pepper, tape (blue paper excluded due to convergence issues)
  - 5 repeats per object-plate combination
- **Diffusion evaluation:**
  - Success vs. initial hand offset (5, 10, 15 cm)
  - Multi-modal grasping (slide & pick vs. direct pick)
  - Failure recovery behavior
- **Combined VLA + diffusion:**
  - Full pick-and-place for all object-plate combinations
  - 5 trials per combination
  - Compare success scores to VLA-only baseline

Evaluation Metric for Pick-and-Place

- Success scored on a 0–1 scale:
    - 1.00: Full task success
    - 0.75: Failed to place on correct plate
    - 0.50: Placed on incorrect plate
    - 0.25: Failed to grasp correct object
    - 0.00: Approached wrong object
- Allows more nuanced comparison than binary success
- Applied to both VLA-only and VLA+diffusion conditions

VLA-Only Performance: Success Rates

- VLA fine-tuned on full grasping and placing for pepper and tape
- Achieved >95% action accuracy in training
- Test success scores (Table I):

End Target	Pepper	Tape
Purple plate	0.45	0.20
Yellow plate	0.40	0.25

- VLA-only can occasionally grasp pepper
- Fails consistently on tape, which requires more precise dexterity
- Scores mainly reflect ability to approach correct object

VLA Reaching Accuracy

- Evaluate offset between hand center and object in x-y plane
- Position recorded when VLA event signal triggers switch
- Compare:
  - VLA with cam1 only
  - VLA with concatenated cam1+cam2
- Results (Fig. 6):
  - Combined cameras: typical offset $\leq$ 6 cm
  - For tape: mean offset reduced from 12 cm to 3 cm
  - Two cameras help compensate for depth ambiguity

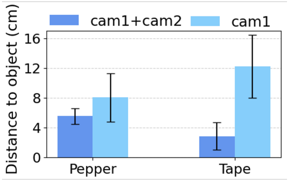

Figure: x-y offset from target object for different camera setups (Fig. 6).

Diffusion Policy: Multi-Modal Grasping

- Objects: tape and blue block
- Training includes:
  - Slide & pick trajectories
  - Direct pick trajectories
  - Equal number of each type
- Test: vary distance from table edge
- Results (Fig. 7):
  - Closer to edge: higher probability of slide & pick
  - Further from edge: higher probability of direct pick
  - All trials (5 per distance) succeed for both objects

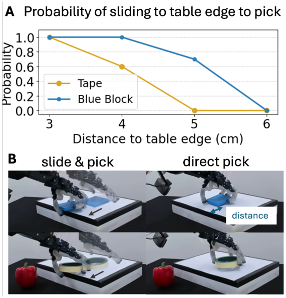

Figure: Multi-modal grasp selection vs. distance to table edge (Fig. 7).

Diffusion Policy: Sensitivity to Initial Offset

- Test grasp success vs. initial hand-object offset
- Offsets in x-y plane: 5, 10, 15 cm
- Objects: pepper and tape
- Results (Fig. 8):
  - Success rate drops below 50% when offset > 15 cm
  - Indicates diffusion requires hand to start near object
  - VLA's typical 6 cm error lies within diffusion's tolerance

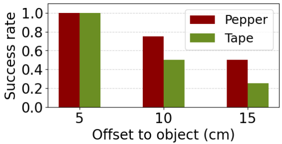

Figure: Grasp success vs. initial offset for diffusion policy (Fig. 8).

Diffusion Policy: Failure Recovery

- Demonstration on picking up paper
- First attempt: incomplete grasp, object slips
- Policy automatically re-attempts grasp
- Event signal behavior (Fig. 9):
  - Remains zero during failed attempt
  - Rises to 1 only when successful lift occurs
- Joint trajectories show multiple grasp attempts

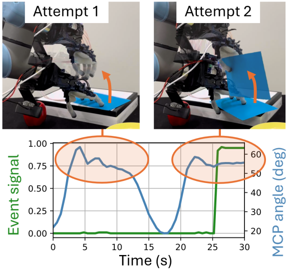

Figure: Diffusion policy recovering from failed grasp attempts (Fig. 9).

Combined VLA + Diffusion: Task Execution

- Full pipeline for pick-and-place (Fig. 10):
  - VLA: approach object and plate
  - Diffusion: grasp and release
  - Event signal σ drives switching
- Trajectories:
  - Vertical motion and σ over time show clear phases
  - x-y trajectories show distinct VLA vs. diffusion segments
- Color coding:
  - Purple: VLA-controlled motion
  - Green: diffusion-controlled motion

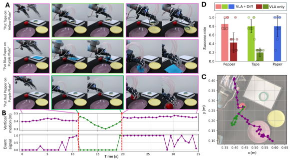

Figure: Combined VLA-diffusion pipeline and success scores (Fig. 10).

Combined VLA + Diffusion: Performance

- Success scores for all object-plate combinations:
  - VLA + diffusion: > 0.8 for all combinations
  - VLA-only (trained on pepper and tape only):
    - Pepper: 0.43
    - Tape: 0.20
- Hybrid approach:
  - More than doubles success compared to VLA-only in many cases
  - Enables robust manipulation of more challenging objects (e.g., tape, paper)
- Demonstrates effective integration of high-level and low-level controllers

Conclusion

- Introduced a **hybrid VLA-diffusion framework** for dexterous manipulation
- First demonstration (to authors' knowledge) of VLA control on a multi-fingered hand
- Key contributions:
  - Event-signal-based switching between VLA and diffusion policies
  - Use of concatenated multi-view images to improve VLA localization
  - Demonstration of multi-modal grasping and failure recovery via diffusion
- Achieved >80% success in language-conditioned pick-and-place
  - Significant improvement over VLA-only baseline

Limitations and Future Work

- **Limitations:**
  - One diffusion policy per object (lookup table selection)
  - openVLA pre-trained only on 1 DoF pinch grippers
  - Limited object set and task diversity
- **Future directions:**
  - More generalized diffusion policies or use of large-scale datasets
  - VLA models trained on diverse multi-fingered hands
  - Incorporation of variable stiffness as a control input
  - Integration of tactile sensing for improved interaction control
- Explore hardware-software co-design to further enhance robustness

Thank you for your attention!

Questions?